SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1520

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**
Question Count: 5
Duration : 60 Minutes

Code of Conduct

I Kunisha Judith. J 192572071 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: KuJ

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- Library pane showing multiple VMs = multiple instances/regions.
- VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.
- Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.
- VM → Manage → Clone = replica/auto-scaling demonstration.

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

1) Introduction :

A hypervisor is software that creates and manages virtual machines (VMs), allowing multiple operating systems to run on the same physical hardware. Hypervisors are classified into two types : Type-1 and Type-2. Both support virtualization, but they differ in architecture, performance, security and management.

Comparison between Type-1 and Type-2 Hypervisor

Criteria	Type-1 Hypervisor	Type-2 Hypervisor
Architecture	Runs directly on physical hardware	Runs on top of a host operating system
Performance	Provides high performance with lower overhead	Has additional overhead due to the host OS
Security	smaller dependency on a general-purpose OS reduces certain attack risks	Security depends on both the host OS and the hypervisor
Manageability	supports centralized management of VMs and resources in datacenters	Generally easier to install and manage on personal computers
Examples	VMware ESXi, Microsoft Hyper-V, Xen.	Oracle VirtualBox, VMware Workstation Pro.

usage	Enterprise servers and cloud datacenters	Development, testing and desktop virtualization

Analysis:

Type-1 hypervisors run directly on physical hardware, reducing the overhead associated with a host operating system. This allows better CPU, memory and I/O performance. They are suitable for mission-critical applications that require high availability, reliability and consistent performance.

Type-1 hypervisors also provide strong isolation between virtual machines and support centralized monitoring, resource allocation and management. However, they still require regular security updates and proper configuration.

Type-2 hypervisors depend on a host operating system. This additional layer can increase resource consumption and introduce security risks if the host OS is compromised. Although they are useful for development and testing, they are generally less suitable for large-scale production datacenters.

conclusion:

A Type-1 hypervisor is the better choice for a cloud datacenter hosting mission-critical applications because it offers lower virtualization overhead, better resource utilization, strong workload isolation, and enterprise level management capabilities. These advantages help maintain application performance, reliability and security.

2) Introduction:

cloud providers often host workloads belonging to multiple organizations on the same physical hardware. For eg: finance systems, healthcare applications and student portals may share a server. Virtualization enables these workloads to operate independently while sharing physical resources.

Virtualization Architecture for Workload Isolation

Virtualization divides a physical server into multiple virtual machines using a hypervisor. Each VM has its own operating system, virtual CPU, memory allocation, storage and virtual network interface.

The finance application can run in one VM, the healthcare application in another, and the student portal in a third. The hypervisor controls access to physical resources and prevents one VM from directly accessing another VM's memory or allocated resources under normal operation.

Isolation can be strengthened through the following measures:

1. VM isolation: separate workloads into different VMs with independent operating systems and controlled resource access.
2. Network segmentation: use virtual networks, VLANs, firewalls and security groups to restrict communication between finance, healthcare and student systems.
3. Access control: Apply role-based access control (RBAC) and least-privilege permissions to limit administrative access.
4. Resource allocation: set CPU, memory, and storage limits to prevent one workload from consuming all available resources.
5. Data protection: Encrypte sensitive data at rest and in transit, with separate access permissions for each workloads.

Two major attack surfaces

1. Hypervisor attack surface:
The hypervisor controls virtual machines and access to physical hardware. A vulnerability in the hypervisor could allow an attacker to escape from one VM and access the host or other VMs. This may expose financial records, healthcare information or student data.

Mitigation: Regularly patch the hypervisor, minimize enabled services, restrict administrative access and monitor suspicious activities.

2. Virtual network attack surface
Virtual switches, virtual network interfaces, and communication between VMs can become attack points. Incorrect network configuration may allow unauthorized access, traffic interception or movement between workloads.

Mitigation: use network segmentation, firewall rules, encrypted communication, intrusion detection and continuous network monitoring.

conclusion:
Virtualization provides workload isolation through separate VMs, controlled resource allocation, and virtual networking. However, isolation can be

weakened by hypervisor vulnerabilities or network misconfigurations. A combination of secure virtualization, network segmentation, encryption and access control is ~~access~~ necessary to protect sensitive workloads.

3)

Introduction:

Virtual machine performance depends on the availability of CPU, memory, storage and network resources. The given utilization snapshots indicates that the host VM may be experiencing resource contention, resulting in slow VM startup and reduced application performance.

Analysis of VM Host utilization:

Metric	observed value	Analysis
CPU utilization	92%	very high utilization, indicating limited processing capacity and CPU contention
memory utilization	88%	High memory usage, potentially causing memory pressure or swapping
Storage I/O Latency	35 ms	Elevated latency may delay disk reads and writes

Average VM boot time	170 secs	slow startup, possibly caused by CPU, memory or storage bottlenecks

Likely Bottlenecks

1. CPU bottlenecks:

CPU utilization of 92% indicates that the host is operating near its processing capacity. When many VMs compete for CPU time, applications may respond slowly and VM startup operations may take longer.

corrective action: Rebalance workloads across hosts, reduce unnecessary VM workloads, allocate virtual CPUs appropriately, and add CPU capacity if the demand remains high.

2. Memory bottlenecks:

Memory utilization of 88% indicates limited available RAM. If the host runs short of physical memory, it may use swapping or ballooning mechanisms, increasing memory-access delays and affecting VM performance.

corrective action: Review VM memory allocation, reduce unnecessary memory consumption, migrate some VMs to other hosts, or increase physical RAM.

3. storage I/O bottlenecks:

storage I/O latency of 35 ms suggests that disk operations may be taking too long. This can result from heavy concurrent read/write operations, slow storage devices, or storage resource contention. VM startup requires reading operating system files and loading services, so storage delays can contribute to the 170-second boot time.

corrective action: Monitor disk IOPS and storage queues, distribute disk-intensive workloads, use faster SSD-based storage where appropriate and improve storage configuration.

4. slow VM boot time:

An average boot time of 170 secs indicates inefficient VM startup. High CPU usage, memory pressure, and storage latency may collectively delay operating system loading and service initialization.

corrective action: Identify unnecessary startup services, investigate storage delays, optimize VM templates, and schedule VM startups in batches rather than starting many VMs simultaneously.

conclusion:

The host shows signs of resource pressure, with high CPU and memory utilization and elevated storage latency. These conditions may be contributing to the long VM boot time. The recommended approach is to monitor resource usage, balance workloads, optimize storage, and increase capacity where necessary. Performance should be measured again after the changes to confirm improvement.

4) Introduction:

A software-Defined datacenter (SDDC) is a datacenter architecture in which compute, storage and networking resources are virtualized and managed through software. Unlike a traditional datacenter, which relies on heavily on manually configured physical infrastructure, an SDDC enables centralized control, automation and flexible resource allocation.

Traditional Datacenter Vs. Software-Defined Datacenter

Criteria	Traditional Datacenter	Software-Defined Datacenter
Resource management	often involves manual configuration	uses centralized software management

compute	Primarily tied to physical server capacity	Uses virtual machines and software-controlled allocation
storage	often depends on dedicated storage hardware	uses virtualized and software-managed storage resources
Networking	Relies heavily on physical network configuration	uses virtual networks and software-defined policies
scalability	May require significant manual hardware deployment	Resources can often be provisioned and adjusted more quickly
Agility	changes can take longer to implement	Automation enables faster deployment and modification

How SDDC improves Agility

1. compute virtualization:

compute virtualization uses a hypervisor to divide physical servers into multiple VMs, CPU and memory resources can be allocated according to workload requirements. Administrators can

create, clone, migrate or remove VMs without installing a separate physical server for each application.

This improves deployment speed and allows resources to be adjusted as demand changes.

2. storage virtualization :

storage virtualization combines storage resources from different physical devices into local storage pools. Administrators can allocate storage to applications based on their capacity and performance needs.

Features such as thin provisioning, snapshots and automated storage management can improve resource utilization, simplify backup operations and reduce the time needed to provision storage.

3. Network virtualization :

Network virtualization creates logical networks independently of the underlying physical network. virtual switches, software-defined networking (SDN), and programmable security policies allow administrators to configure connectivity and segmentation through software.

For example, a new application can be placed on an isolated virtual network without requiring extensive physical network changes. This improves deployment speed and security management.

4. Automation and centralized management

An SDDC integrates the management of compute, storage and networking resources. Automation can support VM provisioning, workload migration, resource scaling and policy enforcement. This reduces repetitive manual work and helps administrators respond quickly to business requirements.

conclusion:

SDDC improves datacenter agility by virtualizing compute, storage and networking and managing them through centralized software. It enables faster provisioning, better resource utilization, easier scalability, and consistent policy enforcement. Therefore, it is more flexible and responsive to changing workload requirements than a traditionally managed datacenter.

3) Introduction :

Multi-cloud federation allows organizations to use services from multiple cloud providers while enabling coordinated access, resource sharing and identity management. However, the project may fail when different providers use incompatible identity formats, authentication methods or authorized authorization policies.

Analysis of the Identity Mismatch :

Identity mismatches occur when cloud providers represent users, roles and permissions differently. For example : one provider may identify a user through an email address, while another uses a unique user ID. similarly a role such as Administrator may have different permissions across providers.

These differences can lead to authentication failures, unauthorized access, incorrect privilege assignments and difficulties managing user accounts. Requiring separate accounts for every provider also increases administrative complexity and the risk of inconsistent access controls.

Proposed secure Identity and privacy design

1. Establish a central identity provider (IdP)

use a trusted identity provider to manage user identities and authentication. The IdP verifies users and issues security tokens or assertions that participating cloud providers can validate.

This reduces the need for users to maintain separate credentials for each cloud provider.

2. Implement Identity federation:

use standard federation protocols such as Security Assertion Markup Language (SAML 2.0) or Open ID Connect (OIDC), depending on provider support. These standards enable participating providers to exchange trusted authentication information.

For example, after a user authenticates through the organization's identity provider, the user can access an authorized service on another cloud without creating an entirely separate identity.

3. standardize identity mapping and authorization
create a common identity-mapping policy to associate users, groups and roles across cloud providers. Map each external identity to the correct local role, and apply role-based access control and least privilege.
For example, a finance analyst should receive access only to the financial resources required for their work, and not administrative access across every cloud.

4. strengthen authentication and access security
use multifaitor authentication (MFA), short-lived tokens, secure token validation and regular permission reviews. Disable access promptly when users leave the organization or change roles. Audit authentication and authorization events across all providers.

5. protect privacy and sensitive information:
share only the identity attributes required for authentication and authorization. Avoid transferring unnecessary personal information between providers. Encrypt data in transit, restrict access to identify logs and establish clear data retention and access policies

6. Monitor and audit federation activity

Maintain centralized visibility into authentication failures, unusual access attempts and privilege changes. Regularly review federation trust relationships, certificates, token settings and provide configurations to identity security weakness.

Conclusion:

The multi-cloud federation project failed because of incompatible identity formats, authentication mechanisms and authorization policies. A secure solution requires a trusted identity provider, standard federation protocols, consistent identity mapping, least-privilege access, and privacy protection. These measures improve interoperability, simplify user access and reduce the risk of unauthorized access across multiple cloud environments.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded :